Why You Shouldn't Use TDOA for Multilateration

Daniel Frisch and **Uwe D. Hanebeck**
Intelligent Sensor-Actuator-Systems Laboratory (ISAS)
Institute for Anthropomatics and Robotics
Karlsruhe Institute of Technology (KIT), Germany
daniel.frisch@kit.edu, uwe.hanebeck@kit.edu

***Abstract*—The maximum likelihood problem arising from multilateration or source localization via signal times of arrival (TOA) leads to a nonlinear least squares problem in target position and target transmission time (TTT). Since we are not interested in the latter, it is usually eliminated from the equation system. We eliminate the TTT in closed form, which is simpler to design and easier to implement than the often used pairwise time differences of arrival (TDOAs). We propose an unweighted nonlinear least squares formulation of the multilateration problem whose optimization with the Levenberg-Marquardt algorithm is very fast.**

***Index Terms*—source localization, multilateration, maximum likelihood, time of arrival (TOA), time difference of arrival (TDOA), Weighted nonlinear least squares, Levenberg-Marquardt**

I. Introduction

Multilateration is a method for passive source localization based on time of arrival (TOA) measurements. It can be realized with a relatively cheap sensor infrastructure and can yield very high accuracy. A major application area is secondary surveillance radar (SSR), i.e., the tracking of cooperative aerial targets.

There are convenient algebraic methods providing a closed-form solution [1], [2], [3]. All these methods, however, introduce some nonlinear transform of the measurement equation, usually to get rid of the square root from the Euclidean distance. This implies, that the solution is no longer optimal under the presence of additive Gaussian measurement noise.

The "gold standard" maximum likelihood estimation requires an iterative search for target position and target transmission time (TTT) via numerical optimization. This is proposed in [4]. Some also require target and sensors being time-synchronized, i.e., the TTT to be known, so that one needs to search for target position only [5], [6].

The TTT can also be eliminated by taking differences of pairs of measurements or by solving a linear least squares problem, which we focus on in this work. Differentiation from prior art in these areas is given in Section III-E and Section IV-C, after having established the notation and problem structure.

II. Problem Description

A. Measurement Equation

A target at unknown location

$$\underline{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \tag{1}$$

transmits a radio signal at unknown time t_0. This signal is received by various sensors at known locations

$$\underline{s}_i = \begin{bmatrix} s_{i,x} \\ s_{i,y} \\ s_{i,z} \end{bmatrix}, \quad \text{for } i \in [1, 2, \ldots, N] \ . \tag{2}$$

Each sensor measures the TOA t_i with zero-mean additive Gaussian noise v_i with stochastic properties

$$\mathrm{Var}\{v_i\} = \sigma_i^2 \ , \qquad \mathrm{Cov}\{v_i, v_j\} = 0 \ \text{ for } i \neq j \ . \tag{3}$$

Therefore, we have the measurement equation

$$t_i = t_0 + \frac{1}{c}\|\underline{x} - \underline{s}_i\| + v_i \ , \tag{4}$$

where c is the signal propagation speed. In vector notation, for all measurements combined, it is

$$\underline{t} = t_0 \cdot \underline{1} + \underline{h}(\underline{x}) + \underline{v} \ , \tag{5}$$

where

$$\underline{t} = \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_N \end{bmatrix}, \quad \underline{h}(\underline{x}) = \frac{1}{c}\begin{bmatrix} \|\underline{x} - \underline{s}_1\| \\ \|\underline{x} - \underline{s}_2\| \\ \vdots \\ \|\underline{x} - \underline{s}_N\| \end{bmatrix}, \quad \underline{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}, \tag{6}$$

with noise covariance

$$\mathrm{Cov}\{\underline{v}\} = \mathbf{C}_v = \begin{bmatrix} \sigma_1^2 & & & 0 \\ & \sigma_2^2 & & \\ & & \ddots & \\ 0 & & & \sigma_N^2 \end{bmatrix} . \tag{7}$$

B. Maximum Likelihood Estimation

The likelihood is therefore

$$f(\underline{t} \mid t_0, \underline{x}) = f_v(t_0 \cdot \underline{1} - \underline{t} + h(\underline{x})) \ , \tag{8}$$

with

$$f_v(\underline{v}) \propto \exp\left\{-\frac{1}{2}\, \underline{v}^\top \mathbf{C}_v^{-1} \underline{v}\right\} . \tag{9}$$

Finally, the maximum likelihood (ML) solution is

$$(\hat{t}_0, \hat{\underline{x}}) = \underset{(t_0, \underline{x})}{\arg\max}\, f(\underline{t} \mid t_0, \underline{x}) \tag{10}$$

$$= \underset{(t_0, \underline{x})}{\arg\min} \|t_0 \cdot \underline{1} - \underline{t} + \underline{h}(\underline{x})\|^2_{\mathbf{C}_v^{-1}} \ . \tag{11}$$

It can, for diagonal $\mathbf{C}_v$, also be written as

$$(\hat{t}_0, \hat{\underline{x}}) = \underset{(t_0,\underline{x})}{\arg\min} \sum_{i=1}^{N} \left[\frac{1}{\sigma_i} \left(t_0 - t_i + \frac{1}{c} \|\underline{x} - \underline{s}_i\| \right) \right]^2 \quad . \tag{12}$$

This nonlinear least squares problem could readily be solved with the Gauß-Newton algorithm [7], [8], or its regularized variant, the the Levenberg-Marquardt algorithm [9], [10], or also Powell's hybrid "dog-leg" method [11], [12]. However, it seems desirable to first get rid of the unknown TTT t_0 that we are not interested in, to not burden the optimizer with unnecessary work.

III. TDOA Method

What is often done is to subtract two TOA measurement equations from each other, yielding the time difference of arrival (TDOA) measurement equation

$$t_i - t_j = \frac{1}{c}\|\underline{x} - \underline{s}_i\| - \frac{1}{c}\|\underline{x} - \underline{s}_j\| + v_i - v_j \quad , \tag{13}$$

which can also be written as

$$\tau_{i,j} = \eta_{i,j}(\underline{x}) + \nu_{i,j} \quad , \tag{14}$$

$$\eta_{i,j}(\underline{x}) = \frac{1}{c}\|\underline{x} - \underline{s}_i\| - \frac{1}{c}\|\underline{x} - \underline{s}_j\| \quad . \tag{15}$$

Two things have changed here: the TTT t_0 got eliminated, as desired, and the measurement noise changed from v_i to

$$\nu_{i,j} = v_i - v_j \quad . \tag{16}$$

To again obtain the appropriate ML estimator, we have to compute the noise covariance of $\nu_{i,j}$

$$\mathrm{Cov}\{\nu_{i,j}, \nu_{k,l}\} = \mathrm{Cov}\{v_i - v_j, v_k - v_l\} \tag{17}$$

$$= \mathrm{E}\{v_i v_k\} - \mathrm{E}\{v_i v_l\} - \mathrm{E}\{v_j v_k\} + \mathrm{E}\{v_j v_l\} \quad . \tag{18}$$

A. Consecutive Topology

Assuming we take differences of consecutive measurements, i.e., $N-1$ measurements $\tau_{i,j}$, in particular,

$$\underline{\tau} = \begin{bmatrix} \tau_{1,2} & \tau_{2,3} & \dots & \tau_{N-1,N} \end{bmatrix}^\top \quad . \tag{19}$$

Then the covariance is

$$\mathbf{C}_\nu = \tag{20}$$

$$\begin{bmatrix} \sigma_1^2 + \sigma_2^2 & -\sigma_2^2 & & & 0 \\ -\sigma_2^2 & \sigma_2^2 + \sigma_3^2 & -\sigma_3^2 & & \\ & -\sigma_3^2 & \sigma_3^2 + \sigma_4^2 & \ddots & \\ & & \ddots & \ddots & -\sigma_{N-1}^2 \\ 0 & & & -\sigma_{N-1}^2 & \sigma_{N-1}^2 + \sigma_N^2 \end{bmatrix} , \tag{21}$$

because

$$\mathrm{Cov}\{\nu_{i,i+1}, \nu_{i,i+1}\} \tag{22}$$

$$= \mathrm{E}\{v_i v_i\} - \mathrm{E}\{v_i v_{i+1}\} - \mathrm{E}\{v_{i+1} v_i\} + \mathrm{E}\{v_{i+1} v_{i+1}\} \tag{23}$$

$$= \sigma_i^2 - 0 - 0 + \sigma_{i+1}^2 \tag{24}$$

$$= \sigma_i^2 + \sigma_{i+1}^2 \quad , \tag{25}$$

and

$$\mathrm{Cov}\{\nu_{i-1,i}, \nu_{i,i+1}\} \tag{26}$$

$$= \mathrm{E}\{v_{i-1} v_i\} - \mathrm{E}\{v_{i-1} v_{i+1}\} - \mathrm{E}\{v_i v_i\} + \mathrm{E}\{v_i v_{i+1}\} \tag{27}$$

$$= 0 - 0 - \sigma_i^2 + 0 \tag{28}$$

$$= -\sigma_i^2 \quad . \tag{29}$$

B. Star Topology

Alternatively, we could use a star-like topology, where we also get $N-1$ measurements $\tau_{i,j}$

$$\underline{\tau} = \begin{bmatrix} \tau_{1,2} & \tau_{1,3} & \dots & \tau_{1,N} \end{bmatrix}^\top \quad . \tag{30}$$

In that case the covariance is

$$\mathbf{C}_\nu = \begin{bmatrix} \sigma_1^2 + \sigma_2^2 & & & \sigma_1^2 \\ & \sigma_1^2 + \sigma_2^3 & & \\ & & \ddots & \\ \sigma_1^2 & & & \sigma_1^2 + \sigma_N^2 \end{bmatrix} , \tag{31}$$

because

$$\mathrm{Cov}\{\nu_{1,i}, \nu_{1,j}\} = \sigma_1^2 + \mathrm{E}\{v_i v_j\} \tag{32}$$

$$= \begin{cases} \sigma_1^2 + \sigma_i^2 \ , & i = j \\ \sigma_1^2 \ , & i \neq j \end{cases} \quad . \tag{33}$$

C. Combinatorial

One may also decide to take all combinations of two measurements. Then $\underline{\tau}$ has

$$\binom{N}{2} = \frac{N \cdot (N-1)}{2} \tag{34}$$

elements, and $\mathbf{C}_\nu$ must be individually compiled from (17) for the respective N and ordering of the combinations.

D. Solver

To obtain the ML solution, we have to solve a weighted nonlinear least squares problem similar to (11) but adapted to the transformed measurements $\tau_{i,j}$

$$\hat{\underline{x}} = \underset{\underline{x}}{\arg\min} \|\eta(\underline{x}) - \underline{\tau}\|^2_{\mathbf{C}_\nu^{-1}} \quad , \tag{35}$$

with $\underline{\eta}(\underline{x})$ the vectorized version of $\eta_{i,j}(\underline{x})$ (15). Now we have a non-diagonally weighted nonlinear least squares problem. Although the Levenberg-Marquardt algorithm can be derived for such problems [13], [14, p. 515], the major implementations do not give the option to provide a weighting matrix. We can, however, rewrite the quadratic form in (35) into a sum of squares [15, p. 6] via the Cholesky decomposition $\mathbf{C}_\nu = \mathbf{R}\mathbf{R}^\top$

$$\left\|\underline{\eta}(\underline{x}) - \underline{\tau}\right\|^2_{\mathbf{C}_\nu^{-1}} \tag{36}$$

$$= \left[\underline{\eta}(\underline{x}) - \underline{\tau}\right]^\top \left(\mathbf{R}\mathbf{R}^\top\right)^{-1} \left[\underline{\eta}(\underline{x}) - \underline{\tau}\right] \tag{37}$$

$$= \left[\underline{\eta}(\underline{x}) - \underline{\tau}\right]^\top \mathbf{R}^{-\top} \mathbf{R}^{-1} \left[\underline{\eta}(\underline{x}) - \underline{\tau}\right] \tag{38}$$

$$= \left[\mathbf{R}^{-1}\left(\underline{\eta}(\underline{x}) - \underline{\tau}\right)\right]^\top \left[\mathbf{R}^{-1}\left(\underline{\eta}(\underline{x}) - \underline{\tau}\right)\right] \quad . \tag{39}$$

Thus, transforming the vector $(\underline{\eta}(\underline{x}) - \underline{\tau})$ with $\mathbf{R}^{-1}$ renders the least squares problem unweighted. However, this matrix multiplication makes computation of the objective function noticeably slower and can be avoided by our proposed method.

E. Distorted Solvers

It appears that some TDOA users do not take into account the correlations between the $\nu_{i,j}$ and simply treat the TDOA measurements as if they had uncorrelated noise, i.e., with $\mathbf{C}_\nu$ diagonal [16, Eq. 29], [17, Eq. 10-11], [18, Eq. 7], [19, Eq. 10]. This may be justified in some cases, when time differences are directly measured, e.g., via cross-correlation of the two received waveforms [20], [21], [22]. But usually, the TOA are determined separately in each sensor, and only subsequently the resulting timestamps being subtracted, yielding the "TDOA" values, which should actually rather be called difference of times of arrival (DOTA) values. Some users, however, do respect the TDOA correlations [23, Eq. 44], [24, Eq. 33], [14, p. 514], [25, Eq. 46], [26, Eq. 3].

IV. Proposed TOA Method

We claim that using differences of TOAs unnecessarily complicates things (in particular, the covariance matrix), and instead propose a different way dealing with the "unwanted unknown" t_0, which does not lead to a non-diagonal weighting matrix.

A. Key Idea

Note that the measurement equation (4) is affine in t_0. Thus, if $\underline{x}$ was given, we could solve for t_0 via linear least squares, i.e., in closed form. And that is exactly what we propose: let the nonlinear solver, just like for TDOA, propose some value for $\underline{x}$ and not for t_0. Then for that specific $\underline{x}$, compute the optimal t_0 and return the resulting loss as objective function.

B. Computing t_0

In particular, we simply solve the vectorized measurement equation (5) for t_0. First, we rearrange it from being affine in t_0 to being linear in t_0

$$\underline{t} - \underline{h}(\underline{x}) = \underline{1} \cdot t_0 + \underline{v} \ . \tag{40}$$

For normally distributed $\underline{v}$ we obtain the ML estimate $\hat{t}_0$ via linear least squares

$$\hat{t}_0 = \left(\underline{1}^\top \mathbf{C}_v^{-1} \underline{1}\right)^{-1} \cdot \underline{1}^\top \mathbf{C}_v^{-1} (\underline{t} - \underline{h}(\underline{x})) \ , \tag{41}$$

which for diagonal $\mathbf{C}_v$ as in (7), becomes

$$\hat{t}_0 = \frac{\sum_{i=1}^N \frac{1}{\sigma_i^2}\left[t_i - \frac{1}{c}\|\underline{x} - \underline{s}_i\|\right]}{\sum_{i=1}^N \frac{1}{\sigma_i^2}} \ , \tag{42}$$

and for $\mathbf{C}_v = \sigma^2 \mathbf{I}$

$$\hat{t}_0 = \frac{1}{N} \sum_{i=1}^N \left[t_i - \frac{1}{c}\|\underline{x} - \underline{s}_i\|\right] \ . \tag{43}$$

Thus, $\hat{t}_0$ is simply the weighted sample mean of $(t_i - \frac{1}{c}\|\underline{x} - \underline{s}_i\|)$. Inserting this into (5) yields a measurement equation that depends only on $\underline{x}$.

C. State-of-Art

A similar approach has already been proposed in [27, Eq. 8+10]. However, they include a penalty that keeps the t_0-estimate of the next iteration close to the current one, which we deem unnecessary and furthermore it requires an initial guess of t_0 which our method does not. But most of all, they treat finding $\hat{t}_0$ and $\hat{x}_0$ as two entirely separate optimizations which are executed alternately. Thus, after each linear least squares $\underline{t}_0$ estimation, they perform a *multi-iteration* nonlinear optimization solving for $\underline{x}$ [27, Alg. 1]. Lastly, they propose a gradient-descent optimization, while we propose, exploiting the problem structure, the Levenberg-Marquardt method for the variant described in Section IV-D, or quasi-Newton for Section IV-E.

Similarly, [28, Eq. 4], [29] use the closed-form $\underline{t}_0$-estimate (42) but solve the entire problem via semidefinite programming, which is generally slower. Also [30, Eq. 6] uses this trick but still arrives at a non-diagonally weighted least squares problem [30, Eq. 7+A7], just like we do for the correctly-weighted TDOA problem, with the somewhat higher computational cost (37), and unlike our diagonally-weighted least squares problem for TOA (45).

D. Levenberg-Marquardt

Inserting (42) into (12) gives a sum-of-squares minimization problem that depends only on $\underline{x}$ and can be solved with the Levenberg-Marquardt algorithm.

Objective Function: Levenberg-Marquardt requires an objective function θ_{LM} that returns a vector of the terms to be squared and summed, which would be

$$\underline{\theta}_{\mathrm{LM}}(\underline{x}) = \begin{bmatrix} \frac{1}{\sigma_1}\left(\hat{t}_0(\underline{x}) - t_1 + \frac{1}{c}\|\underline{x} - \underline{s}_1\|\right) \\ \frac{1}{\sigma_2}\left(\hat{t}_0(\underline{x}) - t_2 + \frac{1}{c}\|\underline{x} - \underline{s}_2\|\right) \\ \vdots \\ \frac{1}{\sigma_3}\left(\hat{t}_0(\underline{x}) - t_3 + \frac{1}{c}\|\underline{x} - \underline{s}_3\|\right) \end{bmatrix} \ , \tag{44}$$

with $\hat{t}_0(\underline{x})$ from (42), where

$$\hat{\underline{x}} = \arg\min_{\underline{x}} [\underline{\theta}_{\mathrm{LM}}(\underline{x})]^\top \mathbf{C}_v^{-1} [\underline{\theta}_{\mathrm{LM}}(\underline{x})] \ . \tag{45}$$

Jacobian: Furthermore, Levenberg-Marquardt needs the Jacobian of $\underline{\theta}_{\mathrm{LM}}(\underline{x})$. First, we define the Jacobian of $\underline{h}(x)$, $\mathbf{J}_{\underline{h}}(\underline{x}) \in \mathbb{R}^{N \times 3}$

$$\mathbf{J}_{\underline{h}} = \frac{1}{c} \begin{bmatrix} (\underline{x} - \underline{s}_1)^\top / \|x - \underline{s}_1\| \\ (\underline{x} - \underline{s}_2)^\top / \|x - \underline{s}_2\| \\ \vdots \\ (\underline{x} - \underline{s}_N)^\top / \|x - \underline{s}_N\| \end{bmatrix} \tag{46}$$

and weight vector $\underline{w}$ containing the diagonal elements of the diagonal $\mathbf{C}_v^{-1}$

$$\underline{w} = \begin{bmatrix} \frac{1}{\sigma_1^2} & \frac{1}{\sigma_2^2} & \cdots & \frac{1}{\sigma_N^2} \end{bmatrix}^\top \ . \tag{47}$$

Then the derivative of $\hat{t}_0(\underline{x})$ from (42), $\frac{\partial \hat{t}_0(\underline{x})}{\partial \underline{x}^\top} \in \mathbb{R}^{1 \times 3}$, is

$$\frac{\partial \hat{t}_0(\underline{x})}{\partial \underline{x}^\top} = -\frac{1}{\underline{1}^\top \underline{w}} \cdot \underline{w}^\top \mathbf{J}_{\underline{h}}(\underline{x}) \ . \tag{48}$$

Method	Weighted	Unweighted
TDOA-Consecutive	229 µs	190 µs
TDOA-Star	228 µs	198 µs
TOA (proposed)	unnecessary	156 µs

TABLE I: Computation times for the setup described in Section V-A.

Method	Weighted	Unweighted
TDOA-Consecutive	0.370	0.452
TDOA-Star	0.370	0.804
TOA (proposed)	–	0.370

TABLE II: Median Euclidean deviation of the results for the setup described in Section V-C.

The desired Jacobian of θ_{LM} from (44), $\mathbf{J}_{\underline{\theta}_{\mathrm{LM}}}(\underline{x}) \in \mathbb{R}^{N\times 3}$, is

$$\mathbf{J}_{\underline{\theta}_{\mathrm{LM}}}(\underline{x}) = \sqrt{\mathrm{diag}(\underline{w})} \cdot \left(\mathbf{J}_{\underline{h}}(\underline{x}) - \frac{1}{\underline{1}^\top \underline{w}} \cdot \underline{w}^\top \mathbf{J}_{\underline{h}}(\underline{x}) \right) \ . \quad (49)$$

E. As Sample Variance Computation

Consider again the nonlinear least squares objective function, slightly modified from (12)

$$\theta(\underline{x}) = \frac{1}{\sum_{i=1}^N \frac{1}{\sigma_i^2}} \sum_{i=1}^N \frac{1}{\sigma_i^2} \left[\frac{1}{c} \|\underline{x} - \underline{s}_i\| - t_i - (-\hat{t}_0(\underline{x})) \right]^2 \quad (50)$$

and compare it to our linear least squares estimate $\hat{t}_0$ from (42)

$$-\hat{t}_0(\underline{x}) = \frac{1}{\sum_{i=1}^N \frac{1}{\sigma_i^2}} \sum_{i=1}^N \frac{1}{\sigma_i^2} \left[\frac{1}{c} \|\underline{x} - \underline{s}_i\| - t_i \right] \ . \quad (51)$$

Inserting (51) into (50) shows that what we have here is precisely a sample variance computation. This has two implications. First, we can also write it as

$$\theta(\underline{x}) = \frac{1}{\sum_{i=1}^N \frac{1}{\sigma_i^2}} \sum_{i=1}^N \frac{1}{\sigma_i^2} \left[\frac{1}{c} \|\underline{x} - \underline{s}_i\| - t_i \right]^2 \quad (52)$$

$$- \left(\frac{1}{\sum_{i=1}^N \frac{1}{\sigma_i^2}} \sum_{i=1}^N \frac{1}{\sigma_i^2} \left[\frac{1}{c} \|\underline{x} - \underline{s}_i\| - t_i \right] \right)^2 \ . \quad (53)$$

Second, we can simply use a readily available, highly optimized, and fast sample variance computation function to compute the weighted variance of the vector

$$\underline{h}(\underline{x}) - \underline{t} = \begin{bmatrix} \frac{1}{c}\|\underline{x} - \underline{s}_1\| - t_1 \\ \frac{1}{c}\|\underline{x} - \underline{s}_2\| - t_2 \\ \vdots \\ \frac{1}{c}\|\underline{x} - \underline{s}_N\| - t_N \end{bmatrix} \ . \quad (54)$$

Say, $\mathrm{SVar}(\underline{y};\ \underline{w})$ computes the weighted sample variance of a vector of univariate samples $\underline{y}$, then our desired objective function can be written as

$$\theta(\underline{x}) = \mathrm{SVar}(\underline{h}(\underline{x}) - \underline{t};\ \underline{w}) \ , \quad (55)$$

Consequently, we can interpret the nonlinear least squares solution

$$\hat{\underline{x}} = \arg\min_{\underline{x}} \theta(\underline{x}) \quad (56)$$

as the $\underline{x}$ that minimizes the sample variance of $(\underline{h}(\underline{x}) - \underline{t})$.

Gradient: The gradient of the objective function is

$$\frac{\partial \theta(\underline{x})}{\partial \underline{x}^\top} = \frac{2}{\underline{1}^\top \underline{w}} \cdot \underline{w}^\top \,\mathrm{diag}(\underline{h}(\underline{x}) - \underline{t})\, \mathbf{J}_{\underline{h}}(\underline{x}) \quad (57)$$

$$- \frac{2}{(\underline{1}^\top \underline{w})^2} \cdot \left(\underline{w}^\top (\underline{h}(\underline{x}) - \underline{t}) \right) \cdot \left(\underline{w}^\top \mathbf{J}_{\underline{h}}(\underline{x}) \right) \ . \quad (58)$$

This can be seen as twice the weighted sample covariance between $(\underline{h}(\underline{x}) - \underline{t})$ and $\mathbf{J}_{\underline{h}}$. But computation with available covariance computation functions would have to be done separately for the three columns of $\mathbf{J}_{\underline{h}}$ such that both samples have the same dimension, one. Either way, with this we can now also solve the three-dimensional TOA multilateration problem with, e.g., a Quasi-Newton algorithm.

V. Evaluation

A. Setup

For evaluation, we place $N = 100$ sensors $\underline{s}_i$ randomly in $[0, 10]^3$, define a ground truth, $t_0 = 0.2$ and $\underline{x} = [3, 1, 5]^\top$, assume $c = 1$, compute the noise-free measurements and add standard normal noise ($\sigma = 1$). We define the initial guess $\underline{x}_0 = [9, 8, 2]^\top$ and optimize in Julia using LeastSquaresOptim.jl, with standard settings (x_tol = f_tol = g_tol = 10^{-8}), using in-place syntax for the objective function and automatic differentiation via ForwardDiff.jl [31] for the Jacobian.

B. Timing

Computation times are shown in Table I. Thus, with the proposed TOA we have a computational load comparable than than unweighted TDOA – or even somewhat lower.

C. Accuracy

Now we randomly vary the sensor locations, repeating the experiment 100,000 times. The root mean square error (RMSE) values are listed in Table II. See Fig. 1 for a visualization of the distribution of the errors. All TDOA methods with the correct weighting matrix as well as the TOA method produce identical results.

D. Interpretation

The proposed TOA method has a runtime comparable than the unweighted TDOA methods – while producing the same result as the correctly weighted TDOA.

VI. Conclusion

We proposed a novel combination of unweighted least-squares objective function and Levenberg-Marquardt solver for the exact maximum likelihood problem in multilateration which is very fast. State-of-the-art methods use differences of pairs of TOAs, called TDOA, but then the least squares problem turns into a weighted one – with non-diagonal covariance matrix, which is easily forgotten in design, error-prone in

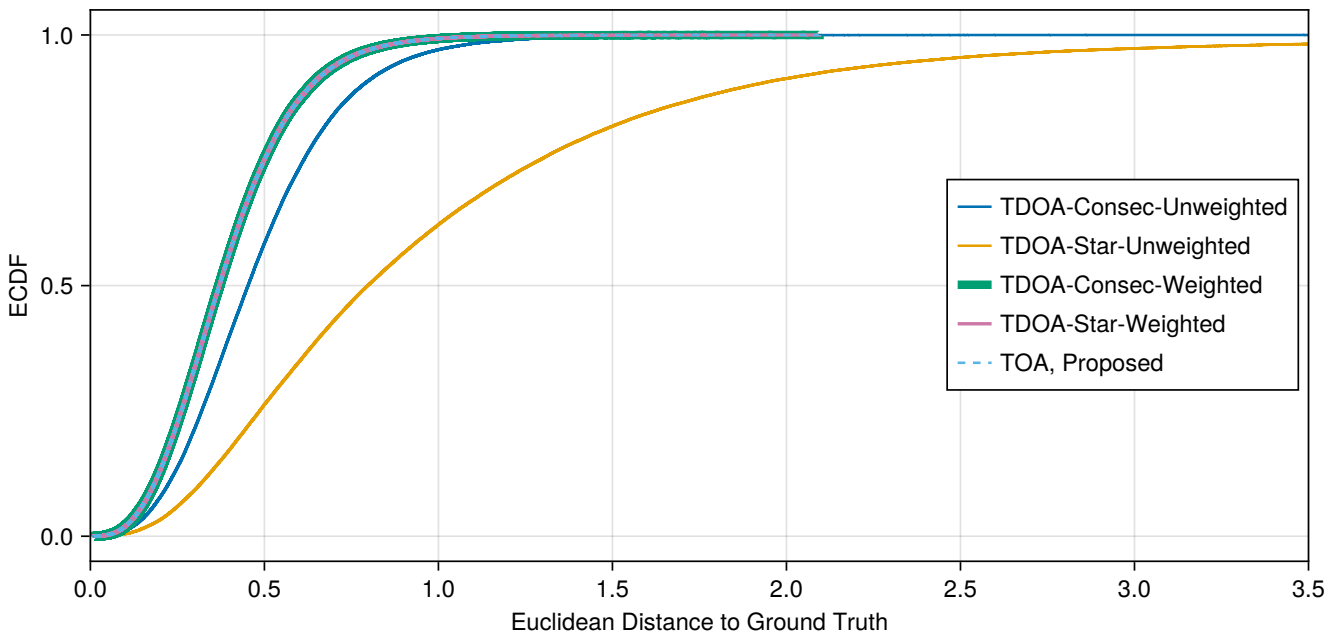

Fig. 1: Evaluation of our proposed TOA multilateration algorithm against four state-of-art methods. The correctly weighted TDOA methods as well as the proposed TOA method identically give the optimal results.

implementation, and slow in runtime. Our method is purely TOA-based, preserves the diagonal weighting matrix, is faster to compute. In particular, we derive a vector-valued objective function that is very simple to implement in-place and is suitable for the Levenberg-Marquardt algorithm (45), as well as a scalar-valued one that can be computed via a a sample variance routine and can be optimized with a Quasi-Newton method (56).

References

[1] Stephen Bancroft. "An Algebraic Solution of the GPS Equations". In: *IEEE Transactions on Aerospace and Electronic Systems* AES-21.1 (1985), pp. 56–59. DOI: 10.1109/TAES.1985.310538.

[2] Abdelmoumen Norrdine. "An algebraic solution to the multilateration problem". In: *Proceedings of the 15th international conference on indoor positioning and indoor navigation, Sydney, Australia*. Vol. 1315. 2012.

[3] Martin Larsson, Viktor Larsson, Kalle Astrom, and Magnus Oskarsson. "Optimal Trilateration Is an Eigenvalue Problem". In: *ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2019, pp. 5586–5590. DOI: 10.1109/ICASSP.2019.8683355.

[4] Fernando Perez-Cruz, Pablo M. Olmos, Michael Minyi Zhang, and Howard Huang. "Probabilistic Time of Arrival Localization". In: *IEEE Signal Processing Letters* 26.11 (Nov. 2019), pp. 1683–1687. ISSN: 1558-2361. DOI: 10.1109/lsp.2019.2944005.

[5] Wenxin Xiong, Christian Schindelhauer, and Hing Cheung So. "Robust time-of-arrival localization via ADMM". In: *Journal of the Franklin Institute* 361.3 (2024), pp. 1582–1599. ISSN: 0016-0032. DOI: https://doi.org/10.1016/j.jfranklin.2024.01.022.

[6] Trung-Kien Le and Nobutaka Ono. "Closed-form and near closed-form solutions for TOA-based joint source and sensor localization". In: *IEEE Transactions on Signal Processing* 64.18 (2016), pp. 4751–4766.

[7] Carl Friedrich Gauss. *Theoria motus corporum coelestium in sectionibus conicis solem ambientium auctore Carolo Friderico Gauss*. sumtibus Frid. Perthes et IH Besser, 1809.

[8] Jorge Nocedal and Stephen J. Wright. *Numerical Optimization*. Second. New York, NY, USA: Springer, 2006.

[9] Kenneth Levenberg. "A Method for the Solution of Certain Non-Linear Problems in Least Squares". In: *Quarterly of Applied Mathematics* 2.2 (1944), pp. 164–168. ISSN: 0033569X, 15524485.

[10] Donald W. Marquardt. "An Algorithm for Least-Squares Estimation of Nonlinear Parameters". In: *Journal of the Society for Industrial and Applied Mathematics* 11.2 (1963), pp. 431–441. DOI: 10.1137/0111030.

[11] M. J. D. Powell. "A Hybrid Method for Nonlinear Equations". In: *Numerical Methods for Nonlinear Algebraic Equations*. Ed. by P. Rabinowitz. dogleg. Gordon and Breach, 1970.

[12] Richard H. Byrd, Robert B. Schnabel, and Gerald A. Shultz. "Approximate solution of the trust region

problem by minimization over two-dimensional subspaces". In: *Mathematical Programming* 40.1 (1988). dogleg, pp. 247–263. ISSN: 1436-4646. DOI: 10.1007/BF01580735.

[13] Henri P Gavin. "The Levenberg-Marquardt algorithm for nonlinear least squares curve-fitting problems". In: ().

[14] C. Mensing and S. Plass. "Positioning Algorithms for Cellular Networks Using TDOA". In: *2006 IEEE International Conference on Acoustics Speech and Signal Processing Proceedings*. Vol. 4. 2006, pp. IV–IV. DOI: 10.1109/ICASSP.2006.1661018.

[15] Daniel Frisch and Uwe D. Hanebeck. "ROTA: Round Trip Times of Arrival for Localization with Unsynchronized Receivers". In: *Proceedings of the 22nd International Conference on Information Fusion (Fusion 2019)*. Ottawa, Canada, July 2019. DOI: 10.23919/FUSION43075.2019.9011167.

[16] Amir Beck, Petre Stoica, and Jian Li. "Exact and Approximate Solutions of Source Localization Problems". In: *IEEE Transactions on Signal Processing* 56.5 (2008), pp. 1770–1778. DOI: 10.1109/TSP.2007.909342.

[17] R. Kaune, C. Steffes, S. Rau, W. Konle, and J. Pagel. "Wide Area Multilateration using ADS-B Transponder Signals". In: *2012 15th International Conference on Information Fusion*. July 2012, pp. 727–734.

[18] Benoit Figuet, Raphael Monstein, and Michael Felux. "Combined Multilateration with Machine Learning for Enhanced Aircraft Localization". In: *MDPI Proceedings* 59.1 (2020). ISSN: 2504-3900. DOI: 10.3390/proceedings2020059002.

[19] Sergei Markochev. "Aircraft Localization Using ATC Data with Nanosecond Precision from Distributed Crowdsourced Receivers". In: *Engineering Proceedings* 13.1 (2021). ISSN: 2673-4591. DOI: 10.3390/engproc2021013012.

[20] Ernst Szabo and Dieter Eier. "Multilateration of Demodulated Radio Signals". In: *2025 Integrated Communications, Navigation and Surveillance Conference (ICNS)*. 2025, pp. 1–9. DOI: 10.1109/ICNS65417.2025.10976931.

[21] Naresh Vankayalapati, Steven Kay, and Quan Ding. "TDOA based direct positioning maximum likelihood estimator and the cramer-rao bound". In: *IEEE Transactions on Aerospace and Electronic Systems* 50.3 (2014), pp. 1616–1635. DOI: 10.1109/TAES.2013.110499.

[22] Noha El Gemayel, Sebastian Koslowski, Friedrich K. Jondral, and Joachim Tschan. "A low cost TDOA localization system: Setup, challenges and results". In: *2013 10th Workshop on Positioning, Navigation and Communication (WPNC)*. 2013, pp. 1–4. DOI: 10.1109/WPNC.2013.6533293.

[23] Y.T. Chan and K.C. Ho. "A simple and efficient estimator for hyperbolic location". In: *IEEE Transactions on Signal Processing* 42.8 (1994), pp. 1905–1915. DOI: 10.1109/78.301830.

[24] Yiu-Tong Chan, H. Yau Chin Hang, and Pak-chung Ching. "Exact and approximate maximum likelihood localization algorithms". In: *IEEE Transactions on Vehicular Technology* 55.1 (2006), pp. 10–16. DOI: 10.1109/TVT.2005.861162.

[25] Yiteng Huang, J. Benesty, G.W. Elko, and R.M. Mersereati. "Real-time passive source localization: a practical linear-correction least-squares approach". In: *IEEE Transactions on Speech and Audio Processing* 9.8 (2001), pp. 943–956. DOI: 10.1109/89.966097.

[26] G. Galati, M. Leonardi, and M. Tosti. "Multilateration (local and Wide area) as a distributed sensor system: Lower bounds of accuracy". In: *2008 European Radar Conference*. 2008, pp. 196–199.

[27] Yuanpeng Chen, Zhiqiang Yao, and Zheng Peng. "A Novel Method for Asynchronous Time-of-Arrival-Based Source Localization: Algorithms, Performance and Complexity". In: *Sensors* 20.12 (2020). ISSN: 1424-8220. DOI: 10.3390/s20123466.

[28] Yanbin Zou and Qun Wan. "Asynchronous Time-of-Arrival-Based Source Localization With Sensor Position Uncertainties". In: *IEEE Communications Letters* 20.9 (2016), pp. 1860–1863. DOI: 10.1109/LCOMM.2016.2589930.

[29] Enyang Xu, Zhi Ding, and Soura Dasgupta. "Source Localization in Wireless Sensor Networks From Signal Time-of-Arrival Measurements". In: *IEEE Transactions on Signal Processing* 59.6 (2011), pp. 2887–2897. DOI: 10.1109/TSP.2011.2116012.

[30] Yanbin Zou, Jingna Fan, Liehu Wu, and Huaping Liu. "Fixed Point Iteration Based Algorithm for Asynchronous TOA-Based Source Localization". In: *Sensors* 22.18 (2022). ISSN: 1424-8220. DOI: 10.3390/s22186871.

[31] J. Revels, M. Lubin, and T. Papamarkou. "Forward-Mode Automatic Differentiation in Julia". In: *arXiv* (2016). ISSN: arXiv:1607.07892 [cs.MS].